RAMAIAH UNIVERSITY OF APPLIED SCIENCES

Faculty of Engineering and Technology
Department of Computer Science and Engineering

Pre-Project Presentation
AirMentor: AI Mentor Intelligence Dashboard for Student Academic Support
Project Batch No. 48

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Introduction

- **Scalability Crisis**: With faculty managing cohorts of 40–120 students, manual performance tracking has become impossible, leading to a significant administrative burden.

- **Intelligence Gap in ERPs**: Current systems store raw data but do not provide predictions, alerts, or student prioritization, leaving faculty without data-driven guidance.

- **Missed Intervention Window**: Struggling students are typically found only after late-stage failures, wasting the critical 5-week window between IA1 and IA2 where proactive mentorship is most effective

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Literature Survey

Ref.	Title & Year	Methodology	Dataset if applicable	Limitations	Observations
[1]	Ahmed, 2024 (Wiley)	SVM, DT, NB, KNN, ANN	Single institution	No deployment; no mentor interface	Good baseline comparison; confirms ML viability
[2]	Frontiers in Education, 2025	XGBoost + SHAP	Multi-institution	Offline only; SHAP shown to researchers, not faculty	Confirms XAI need; gap in actionability identified
[3]	MDPI – CNN + BiGRU, 2025	CNN + BiGRU Hybrid	University LMS logs	Black-box; computationally heavy	Deep learning viable but not faculty-friendly
[4]	EASE-Predict, Liu et al., 2025	Stacking Ensemble (RF+GB+SVM)	MOOC clickstream	77.4% AUC; no real deployment	Ensemble approach validated; deployment gap confirmed
[5]	Nature Sci. Reports, 2025	XGBoost + LIME/SHAP	10,000 students	Offline analysis onlyStrong e	Strong XAI paper; no production use

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Literature Survey

Ref.	Title & Year	Methodology	Dataset if applicable	Limitations	Observations
[6]	GAR EWS – UOC, 2024	Continuous EWS + nudges	LMS engagement logs	Requires clickstream — unavailable in Indian ERPs	Best deployed system; data requirement is the gap
[7]	XGBoost EWS – Mexico, 2025	XGBoost + SITE integration	40,000 pre-enrollment	Pre-enrollment data only	Good production model; Indian ERP data not compatible
[8]	AutoML + XAI Dashboard, 2024	AutoML + SHAP + DICE	MOOC data	Not live; no class imbalance handling	DICE counterfactuals idea validated
[9]	SAPPNet, Nature, 2024	Deep spatial neural network	COVID survey data	Requires psychology questionnaire	Interesting features; not ERP-native
[10]	IJECE, 2024 – Indian Hybrid MLp	Hybrid ML, multi-phase	Indian college ERP	Offline only; no actionable output	Closest to our context; confirms Indian data constraint gap

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Literature Survey

Gaps in the existing/research works

1. Although multiple ML models achieve high accuracy for student performance prediction, most studies produce outputs only interpretable by researchers not actionable by faculty in daily teaching workflows.
2. Despite growing interest in Educational Data Mining, no reviewed system has been deployed in production for daily use at an Indian private engineering college using only ERP-native data (attendance + IA marks + CGPA).
3. Existing EWS predict semester-level CGPA range; no prior work predicts per-course failure probability (e.g., P(fail Operating Systems) = 74%) after IA1, which is the granularity mentors need.
4. Graph-based methods such as Louvain community detection and PageRank have not been applied to generate ranked daily mentor priority lists from student sections.

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Problem Statement

- Current academic support remains reactive, identifying struggling students only after late-stage failures rather than utilising the critical 5-week intervention window between IA1 and IA2.
- To address this, there is a need for a production-grade ML system that converts ERP data into daily, actionable intelligence by predicting per-course failure risk early.
- This system empowers faculty with prioritised daily action lists and SHAP-based explanations to improve student outcomes through proactive mentorship

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Objectives

1. **Objective 1:** To identify students at risk of course failure after IA1 using a 28-feature ML ensemble (XGBoost + LSTM + TabNet), targeting ≥87% AUC-ROC.
2. **Objective 2:** To develop a graph-based section intelligence layer using Louvain community detection and PageRank to generate daily 3-tier mentor priority lists.
3. **Objective 3:** To analyse per-course risk predictions using SHAP explainability and counterfactual reasoning, translating model outputs into plain-English faculty-readable reason cards.
4. **Objective 4:** To deploy a production-grade, three-role dashboard (Course Leader / Mentor / HoD) on the RUAS server with daily automated email alerts and intervention outcome logging

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Methods /Methodology

Objective	Method	Justification
Per-course Risk Prediction	Multi-model Ensemble: **XGBoost + LSTM + TabNet**using 28 engineered features and **SMOTE** for class imbalance .	XGBoost handles tabular data; LSTM captures temporal patterns; TabNet models multi-course interactions .
Section Prioritisation	**Louvain Community Detection** + **PageRank** on a student risk-feature similarity graph .	Provides a ranking system to identify the top 5 daily priorities for mentors among 40-50 students.
Explainability	**TreeSHAP** for plain-English reason cards and **Counterfactual Reasoning** (DICE) .	Transforms "black-box" ML scores into actionable intelligence by explaining *why* a student is at risk.
Production Deployment	**FastAPI + Streamlit + PostgreSQL + Redis + Docker** on university Linux server.	Enables real-time batch processing of 3,000+ student-course records every morning at 6 AM.

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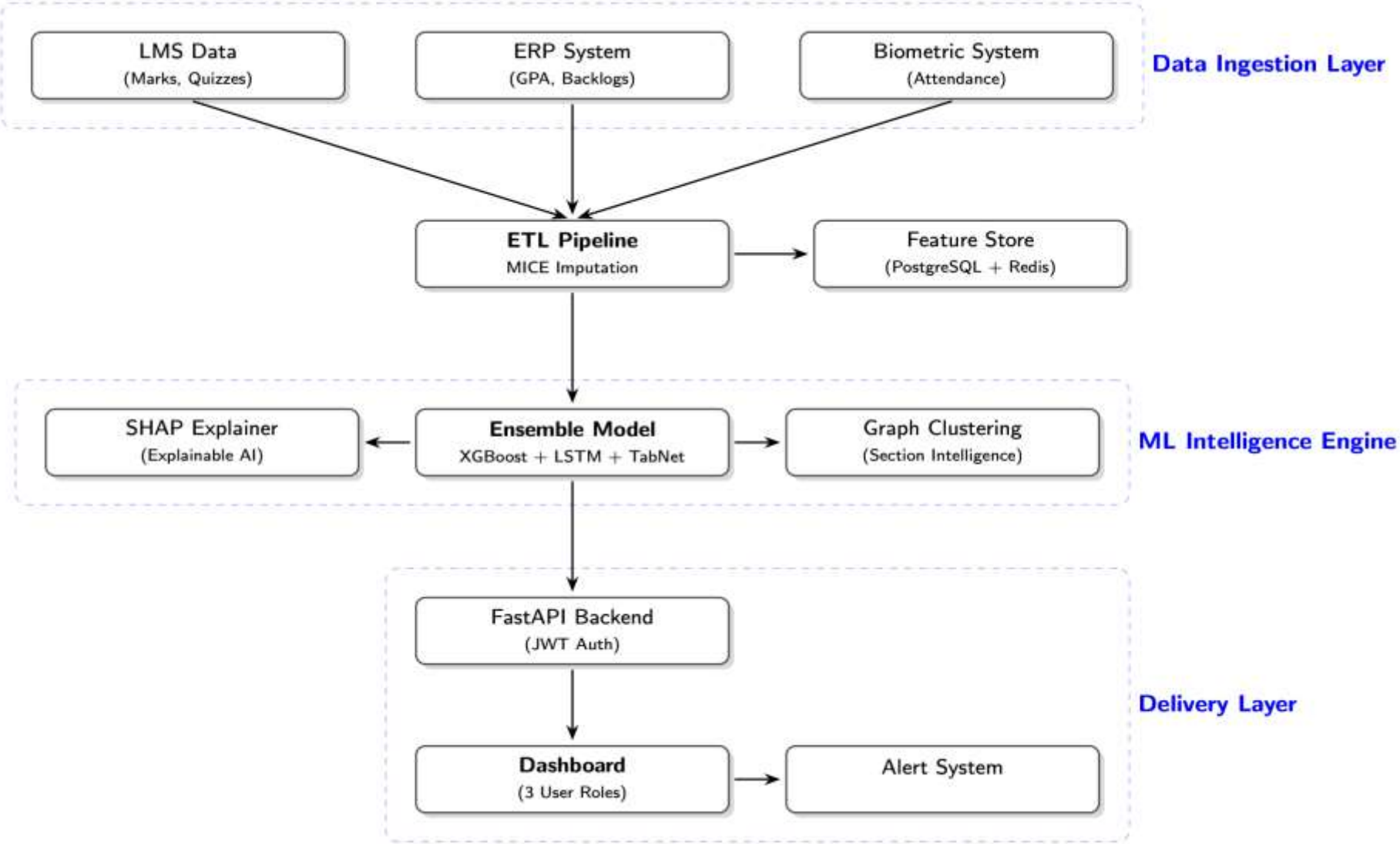
System Architecture



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Expected Outcomes

- Functional Dashboard: Per-course failure probability per student after IA1 · Daily 8AM mentor priority digest · SHAP reason cards · Three-role dashboard deployed on RUAS server
- Performance: ≥87% AUC-ROC, 82% recall on high-risk students, Dashboard response <2 seconds, 3,000 predictions processed in <15 minutes daily
- Practical Impact: 15% reduction in student backlogs, 2+ hours/week saved per course leader, First production-deployed mentor-centric EWS at an Indian private engineering college
- Research Contribution: Novel application of SHAP-to-language translation and graph clustering for faculty decision support

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Sustainable Development Goals(SDG) Aligned to your project

- **SDG 4 – Quality Education (Primary) Target 4.1 & 4.4:** Early identification and proactive support reduce backlogs by 15%, directly improving course completion and quality of tertiary education.

- **SDG 8 – Decent Work & Economic Growth (Secondary) Target 8.6:** Reduced backlogs lead to faster graduation and earlier workforce entry, improving employment outcomes for students.

- **SDG 10 – Reduced Inequalities (Tertiary) Target 10.2:** SHAP transparency and fairness audit ensure equitable support for lateral entry students (22% higher baseline risk) receive proportional attention without algorithmic bias.

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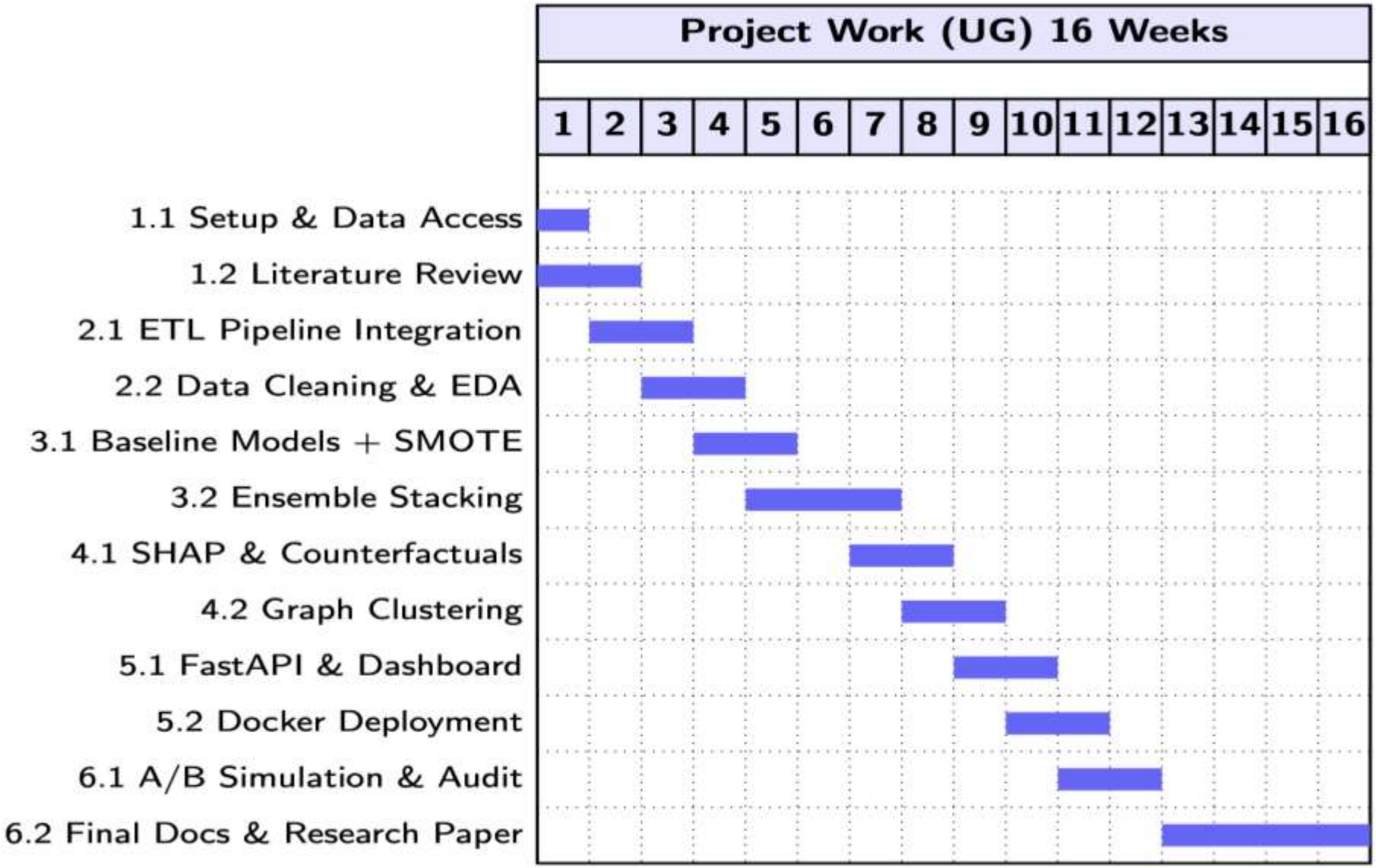
Project Timeline



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Conclusion

- **Integrated AI Pipeline:** AirMentor addresses production-deployment gaps by implementing a four-layer AI pipeline that utilises an ensemble of **XGBoost, LSTM, and TabNet** to provide per-course risk modelling.

- **Faculty-Centric Intelligence:** By combining SHAP explainability with graph-based clustering, the system delivers prioritised daily actions through a role-based dashboard, saving faculty over 2 hours per week.

- **Quantifiable Impact:** Successful implementation targets a **1**5% reduction in backlogs within the critical 5-week intervention window, fulfilling UN SDG 4 objectives and providing a research-grade contribution to engineering education.

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Thank You

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